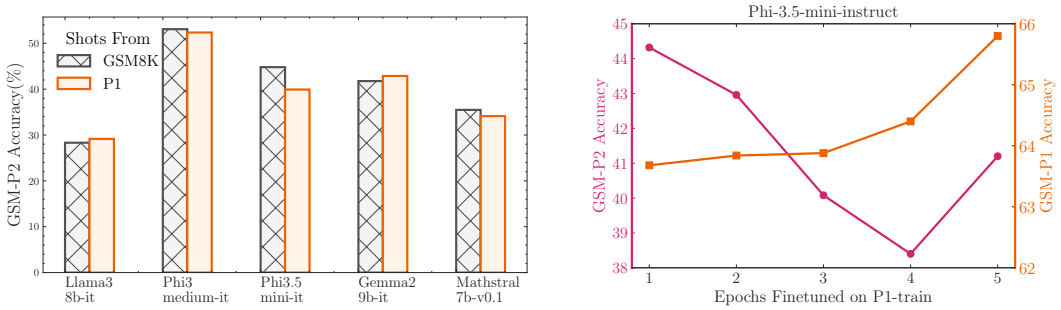

(a) (b)

Figure 11: Using in-context shots or finetuning on `GSM-P1` does not improve performance on `GSM-P2`: (a) Compared to the case where 8 shots come from `GSM8K`, when we include shots from `GSM-P1`the performance on `GSM-P2` does not improve. (b) Finetuning on `GSM-P1` can improve performance on `GSM-P1` but not on `GSM-P2`.

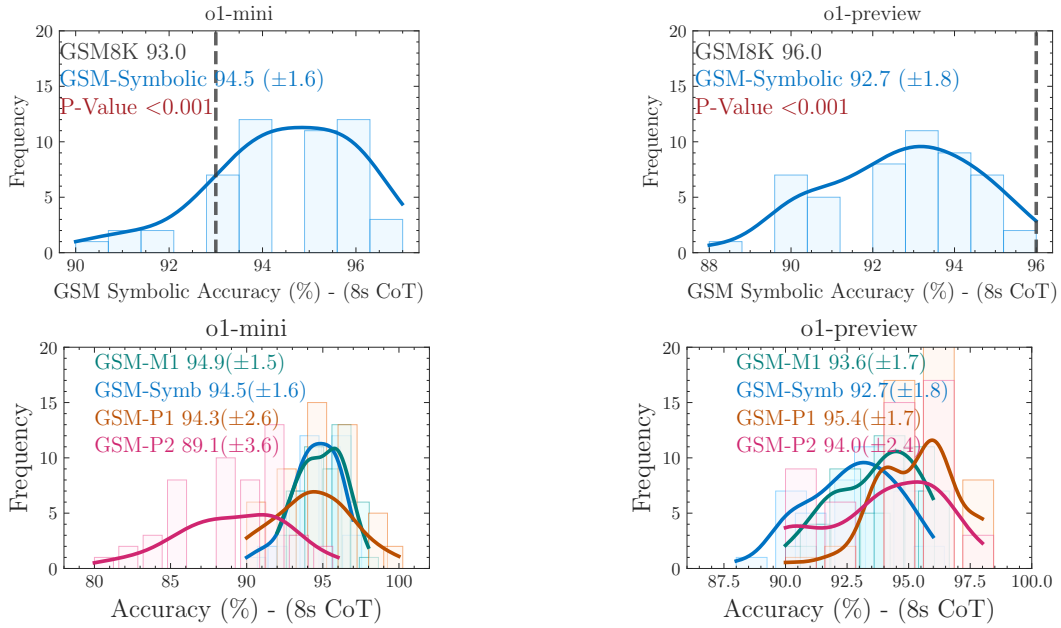

Figure 12: Results on o1-mini and o1-preview: both models mostly follow the same trend we presented in the main text. However, o1-preview shows very strong results on all levels of difficulty as all distributions are close to each other.

o1-preview struggles with understanding mathematical concepts, naively applying the 10% inflation discussed in the question, despite it being irrelevant since the prices pertain to this year. Additionally, in Fig. 14, we present another example highlighting this issue.

Overall, while o1-preview and o1-mini exhibit significantly stronger results compared to current open models—potentially due to improved training data and post-training procedures—they still share similar limitations with the open models.

GSM-NoOp

Liam wants to buy some school supplies. He buys 24 erasers that now cost $6.75 each, 10 notebooks that now cost $11.0 each, and a ream of bond paper that now costs $19. How much should Liam pay now, assuming that due to inflation, prices were 10% cheaper last year?

---

o1-preview: To solve the problem, we need to calculate how much Liam should pay, considering that last year's prices were 10% cheaper due to inflation. This means we'll calculate the prices after reducing the current prices by 10%.

Step 1: Calculate last year's prices by reducing the current prices by 10%.
- Eraser price last year: $\$6.75 \times 0.90 = \$6.075$
- Notebook price last year: $\$11.00 \times 0.90 = \$9.90$
- Bond paper price last year: $\$19.00 \times 0.90 = \$17.10$

Step 2: Calculate the total cost using last year's prices.
- Total cost for erasers: 24 erasers $\times \$6.075 = \$145.80$
- Total cost for notebooks: 10 notebooks $\times \$9.90 = \$99.00$
- Total cost for bond paper: 1 ream $\times \$17.10 = \$17.10$

Step 3: Sum up all the costs to get the total amount Liam should pay.
$\$145.80 + \$99.00 + \$17.10 = \$261.90$

Answer:$261.90

Figure 13: Sample response from o1-preview on an example from GSM-NoOp: the model blindly applies the inflation rate, even though the inflation amount is irrelevant as the question clearly indicates the given prices are for "now" and not last year.

GSM-NoOp

A loaf of sourdough at the cafe costs $9. Muffins cost $3 each. If we purchase 10 loaves of sourdough and 10 muffins, how much more do the sourdough loaves cost compared to the muffins, if we plan to donate 3 loaves of sourdough and 2 muffins from this purchase?

---

o1-preview:
To solve this problem, we need to compare the total cost of the sourdough loaves and muffins after considering the donated items.

Step 1: Calculate the number of items kept after donations.
- Sourdough loaves kept: 10 loaves $- 3$ donated $= 7$ loaves
- Muffins kept: 10 muffins $- 2$ donated $= 8$ muffins

Step 2: Calculate the total cost of the items kept.
- Total cost of sourdough loaves: 7 loaves $\times \$9 = \$63$
- Total cost of muffins: 8 muffins $\times \$3 = \$24$

Step 3: Find the difference in total cost.
- Difference: $\$63 - \$24 = \$39$

Answer:$39

o1-mini:
Let's break down the costs step by step to determine how much more the sourdough loaves cost compared to the muffins after accounting for the donations.

1. Initial Purchase Costs
Sourdough Loaves:
Price per loaf: $9
Quantity purchased: 10
Total Cost: 10 loaves * $9 = $90
Muffins:
Price per muffin: $3
Quantity purchased: 10
Total Cost: 10 muffins * $3 = $30
2. Donated Items and Their Value
Donated Sourdough Loaves:

Quantity donated: 3
Value: 3 loaves * $9 = $27
Donated Muffins:

Quantity donated: 2
Value: 2 muffins * $3 = $6
3. Net Costs After Donation
To find the effective cost of what you're actually keeping (i.e., not donating):

Sourdough Loaves:
Quantity kept: 10 purchased - 3 donated = 7
Net Cost: $90 (total) - $27 (donated) = $63

Muffins:
Quantity kept: 10 purchased - 2 donated = 8
Net Cost: $30 (total) - $6 (donated) = $24

4. Difference in Net Costs
Sourdough Loaves: $63
Muffins: $24
Difference: $63 - $24 = $39

Conclusion
After accounting for the donations, the sourdough loaves cost $39 more than the muffins.

Figure 14: Sample response from o1-preview and o1-mini on an example from GSM-NoOp: while the donation amount is irrelevant to the price difference, the models subtract the amount we donate.